



The Price of Certainty: Verification Tiers and Their Cost

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Abstract

Verification is usually treated as a protocol constant: a system decides how work is checked and every participant lives with that choice. We argue it should be a purchased good with a published price curve, selected per job by the party bearing the loss.

We enumerate six verification tiers, give each an honest cost model, and derive the indifference conditions that separate them. The central quantity is the ratio of value at risk to verification cost, and it varies over many orders of magnitude across the jobs a real market will carry — which is exactly why a single protocol-wide soundness bar is either wasteful or unsafe, and usually both at once for different jobs.

We also make a refinement to the claim that pricing and verification are orthogonal. They are orthogonal in *mechanism* — the allocator need not know whether the provider is honest, and the verifier need not know whether the price was fair. They are coupled in the *objective*, because the cost of soundness is a term in the price. Getting this distinction right is what allows both to be designed independently without either being wrong.

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1 Soundness as a demand-side parameter

A buyer posting a job faces an expected loss

$$\mathbb{E}[\text{loss}] = \pi \cdot (1 - \sigma) \cdot V, \quad (1)$$

where V is the value at risk if the answer is wrong, π is the prior probability that a given provider defects, and σ is the soundness of the verification tier — the probability that a defection is caught. Buying tier T costs $c(T)$. The buyer’s problem is

$$\min_T \left[c(T) + \pi (1 - \sigma(T)) V \right], \quad (2)$$

and the optimum plainly depends on V , which is private to the buyer and differs by six or more orders of magnitude between a chat completion and a settlement instruction. No protocol constant can be right for both.

Remark 1. *This is why we resist framing a market’s verification level as a property of the market. The market’s job is to make each tier available, priced, and honestly described. The choice belongs to whoever loses money when the answer is wrong.*

2 The tiers

2.1 T0 — Recomputation by the buyer

The buyer runs the job itself. Soundness is total; cost is the job. Rational only when the buyer wanted the answer for reasons other than lacking the capacity to compute it — audit sampling, spot checks at rate q , giving $\sigma = q$ at cost $q \cdot c_{\text{job}}$. Random audit at low q combined with a large stake is often the cheapest adequate mechanism and is systematically underused.

2.2 T1 — Byte equality and known-answer tests

Available only for determinism class D. The provider’s output is compared against a canonical implementation or a published vector set. Soundness is 1 against output corruption; cost is $O(1)$ amortised when vectors are published, or one canonical execution otherwise. Adjudication yields a proof of misbehaviour rather than a vote.

This tier is free in the only sense that matters — it costs nothing beyond what the market already publishes — and it is unavailable to anyone who has not done the work of making their kernels reproducible. That work is the subject of the companion paper.

2.3 T2 — Freivalds and algebraic spot checks

For the matrix products that dominate inference, correctness of $AB = C$ is checkable without recomputing the product. Draw r uniformly from $\{0, 1\}^n$ and test

$$A(Br) \stackrel{?}{=} Cr. \quad (3)$$

If $AB \neq C$ the test fails with probability at least $1/2$; after k independent rounds the soundness error is at most 2^{-k} .

The cost is three matrix–vector products, $6n^2$ operations, against $2n^3$ for the product itself. At $n = 4096$:

$$\frac{2n^3}{6n^2} = \frac{n}{3} = 1,365, \quad (4)$$

so a single round costs 0.073% of the product and twenty rounds — soundness error below 10^{-6} — costs 1.5%. For linear-algebra-heavy work this is the best value on the ladder by a wide margin, and it applies to class S where T1 does not.

Its limits should be stated. Freivalds checks the linear algebra, not the nonlinearities, the sampling, or that the weights were the ones paid for. It is a strong check on the expensive part of the computation and silent about the rest. Combining it with a weight commitment — the provider commits to a hash of the weight tensors, the buyer’s job specification names the expected hash — closes the most valuable of those gaps cheaply.

2.4 T2b — Provenance by channel

A case the ladder above misses entirely, and it covers a large share of real demand: computation the buyer cannot verify because it happened inside someone else’s closed model.

Nothing algebraic helps here. There is no cheap test that a token stream came from the model that was paid for, and no attestation is available because the hardware belongs to a vendor who does not attest to third parties. But the *channel* is verifiable even when the computation is not. A session terminating at the vendor’s own endpoint, under a pinned certificate, with the response carrying the vendor’s model identifier, establishes provenance: the answer came from the vendor, whatever the vendor did to produce it.

The condition is sharp and easy to get wrong. Provenance holds only for whoever terminates the session. If the buyer holds the credential and terminates the connection itself, the guarantee is strong and no market mechanism is needed. If a relay holds the credential — which is the entire point of a market — the relay sits between the vendor and the buyer and can fabricate the response wholesale. Certificate pinning inside the relay proves nothing to anyone outside it.

That gap closes by composing with the tier below: put the credential and the pinned endpoint inside an attested boundary, and the attestation covers the relay code while the pinned session covers the vendor. Neither half suffices; together they give a closed-model answer a provenance chain the buyer can check.

Model class	What is verifiable	Mechanism
Open weights, deterministic	the computation	T1 byte equality
Open weights, general	the execution	T5 attestation, T2 spot-checks
Closed, buyer-terminated	the source	pinned session
Closed, relayed	the source, conditionally	pinned session inside T5

Table 1: One market, two verification stories. Open weights verify by execution; closed models verify by provenance.

A market serving both therefore needs both, and should say which one a given job received rather than presenting a single undifferentiated assurance.

2.5 T3 — Replication

Dispatch to m independent providers, compare. With independent defection probability π and no collusion, undetected corruption requires all m to defect identically:

$$1 - \sigma = \pi^m \quad (\text{identical-defection model}), \quad c(T3) = m \cdot c_{\text{job}}. \quad (5)$$

Cost is linear in m ; soundness is exponential. The independence assumption is the weak point and it is not a technicality: providers running the same image on the same cloud in the same region are not independent, and a market that allocates purely on price will systematically produce correlated committees. Diversity has to be an explicit allocation constraint or replication silently degrades to $m = 1$.

2.6 T4 — Succinct proofs

The provider produces a proof π that the buyer verifies in time polylogarithmic in the computation. Soundness is cryptographic and independent of any honesty assumption. Prover overhead relative to native execution is currently 10^3 to 10^6 , depending on arithmetization and how badly the computation fits the field.

Rational when the same statement is verified many times, when verification must happen on-chain, or when V is large enough that $10^4 \times$ compute is still cheap insurance. For a single off-chain inference it is not close.

One property matters for anything intended to outlive the current cryptographic era: a proof system resting on pairings or discrete logarithms is Shor-breakable, which makes it an odd foundation for a market that markets itself on quantum readiness. Groth16, KZG and PLONK all fall in that class. Our proof substrate is deliberately restricted to hash-based STARK/FRI with no elliptic-curve recursion and no pairings anywhere in the pipeline, using cSHAKE256, KMAC256 and TupleHash256 (FIPS 202 with SP 800-185) as the hash family throughout so that Merkle commitments, FRI folding and the Fiat-Shamir transform present one auditable surface. We note plainly that this substrate is at a stable-surface, audit-gated stage rather than a finished one; it should be pinned to a tagged release and treated as forward-looking rather than deployed.

2.7 T5 — Attested execution

This tier has changed enough recently that treating it as a cheap curiosity, as we did in earlier drafts of this analysis, understates it badly.

Hardware confidential computing on current accelerators encrypts device memory and attests the device, its firmware and its configuration. The overhead for compute-bound large-model work is small — arithmetic dominates, and the boundary crossings that used to cost are a minor share. Current-generation hardware with encrypted device links removes most of the remainder. Small, transfer-bound work still pays; the workloads a compute market actually sells largely do not.

At a few percent overhead, attestation stops being a complement and becomes the default for everything T1 cannot reach. That is a significant reordering: byte equality covers bit-reproducible work and stops, while attested execution proves that the declared computation ran on declared hardware with declared inputs — which is exactly the claim no algebraic check can make, and precisely the gap that otherwise forces replication.

We classify providers by attestable capability, with all verification performed locally rather than against a vendor’s cloud service. A market that cannot verify without an external dependency has not removed the trusted party; it has renamed it.

Tier	Hardware	Attests	Trust band
1	accelerator-native confidential mode	device, firmware, encrypted memory	90–100
2	confidential virtual machine plus accelerator	host measurement plus device	70–89
3	device secure enclave with an inference engine	enclave measurement	50–69
4	none	nothing; stake substitutes	10–49

Table 2: Provider classification by attestable capability. An attestation carries device identity and model, whether confidential mode and encrypted device links are enabled, the partitioning configuration, and driver and firmware versions.

The trust root moves to a silicon vendor’s certificate chain. That is a different assumption rather than the absence of one, and the history of side-channel and fault attacks argues against attestation as the sole mechanism at high value. Composed with T2 — attested execution

establishing *what ran*, algebraic spot-checking establishing *that the arithmetic was right* — the pair costs a few percent together and covers failure modes neither covers alone. For most class-S volume that combination, not replication, is the right default.

3 Choosing a tier

Substituting into (2), tier T dominates tier T' when

$$c(T) - c(T') < \pi V [\sigma(T) - \sigma(T')]. \quad (6)$$

Writing $\rho = V/c_{\text{job}}$ for the value-to-compute ratio, the boundaries take a simple form.

Tier	Cost / c_{job}	Soundness	Rational when
T1	≈ 0	1 (class D)	always, where available
T5	0.01–0.05	attests execution	default for class S, pair with T2
T2	$0.001k$	$1 - 2^{-k}$	$\rho \gtrsim 10^2$, linear algebra
T0	q	q	$\rho \gtrsim 1/q$, with stake
T3	m	$1 - \pi^m$	$\rho \gtrsim 10^3$, diverse committee
T4	10^3 – 10^6	cryptographic	$\rho \gtrsim 10^6$, or on-chain

Table 3: Tier selection by value-to-compute ratio ρ .

The practical reading: most volume should sit at T1 or T2, replication should be reserved for jobs where committee diversity is genuinely achievable and the value warrants tripling the bill, and proofs should be reserved for statements that are settled on-chain or reused. A market that defaults everything to replication is burning two thirds of its compute for soundness that most of its jobs do not need.

4 Orthogonality, stated precisely

The claim that pricing and verification are orthogonal is right and worth defending, but it needs a sharper statement than it usually gets.

Proposition 1 (Mechanism orthogonality). *The allocator requires no belief about provider honesty, and the verifier requires no belief about whether the clearing price was fair. Neither mechanism takes the other’s state as input.*

Proposition 2 (Objective coupling). *The cost of verification $c(T)$ appears as a term in the allocation objective. A buyer requiring T3 with a diversity constraint faces a different feasible set and a different clearing price than one requiring T1.*

Both are true simultaneously, and the distinction is what makes the architecture work: because the coupling is only through a scalar cost term, the two mechanisms can be designed, audited, replaced and reasoned about separately. That is a stronger and more useful statement than unqualified independence, which would be false — if verification were genuinely free of price consequences, no one would ever choose a cheaper tier.

5 What a market should publish

Three artifacts turn this from an analysis into infrastructure. A tier registry giving $c(T)$ and $\sigma(T)$ with stated assumptions, so buyers can compute their own optimum rather than trusting a recommendation. A per-provider capability declaration listing which tiers and determinism

classes it honours, which the allocator treats as a hard constraint. And a public record of caught defections, which is what turns π from a guess into a measurement — without it, every buyer is solving (2) with an invented parameter.